

Annex 5

WHO listing of quality control laboratories: procedure for assessing the acceptability, in principle, of quality control laboratories for use by United Nations agencies

Update of *Prequalification of quality control laboratories: procedure for assessing the acceptability, in principle, of quality control laboratories for use by United Nations agencies*, WHO Technical Report Series No. 1003, 2017, Annex 3.

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Contents

Abbreviations	170
1. Introduction	170
2. Steps of the procedure	172
2.1 Publication of invitation for expressions of interest	172
2.2 Submission of expressions of interest and laboratory information	173
2.3 Screening of submitted laboratory information	174
2.4 Evaluation of the laboratory information	174
2.5 Site audit	175
2.6 Report and outcome of audit	175
2.7 Results of assessment	176
2.8 Monitoring of WLQCLs	177
2.9 Monitoring of complaints	178
2.10 Cost recovery	179
2.11 Confidentiality undertaking	179
2.12 Conflict of interest	179
References	181

Abbreviations

QCL	quality control laboratory
WHO	World Health Organization
WLA	WHO-listed authority
WLQCL	WHO-listed quality control laboratory

1. Introduction

This document provides a second revision of the procedure originally published as Annex 12 in World Health Organization (WHO) Technical Report Series No. 961, 2011, and revised for the first time in Annex 3 of WHO Technical Report Series No. 1003, 2017. In this second revision, the procedure formerly referred to as “prequalification of quality control laboratories” is now designated “WHO listing of quality control laboratories”, ensuring alignment with WHO’s broader regulatory terminology and listing mechanisms, such as the WHO-listed authorities (WLAs). Unlike product-focused prequalification, the listing model better reflects the role of laboratories as service providers, acknowledges evaluations already conducted under the WLA framework and reduces duplication of assessment.

WHO provides United Nations agencies and their partners, national authorities of WHO Member States, and procurement agencies serving national authorities and United Nations agencies, on request, with advice on the acceptability, in principle, of quality control laboratories (QCLs) that are found to meet WHO-recommended quality standards for such laboratories. These standards are set out in *WHO good practices for pharmaceutical quality control laboratories* (1) and include, where applicable, good practices for pharmaceutical microbiology laboratories (2), as well as the relevant parts of good manufacturing practices (3) and of the *WHO guideline on the implementation of quality management systems for national regulatory authorities* (4). This is done through a standardized assessment procedure. The purpose of the assessment procedure is to evaluate whether the QCLs to be used for the quality control of pharmaceutical products meet the requirements recommended by WHO for such laboratories.

Participation in the procedure for designation as a WHO-listed quality control laboratory (WLQCL) is voluntary, and any pharmaceutical QCL (governmental or private) providing quality control services for pharmaceutical products to United Nations agencies and their partners, national authorities of WHO Member States, and procurement agencies serving national authorities and United Nations agencies is eligible.

Accreditation, for example under ISO/IEC 17025, is encouraged. While accreditation is not a mandatory prerequisite, laboratories are recommended to work towards obtaining accreditation, as it demonstrates the implementation of a quality management system and supports their efforts towards the WHO listing process. Moreover, accreditation involves periodic surveillance and reassessment, which supports the sustainability of quality management within the laboratory and drives continuous improvement, both of which are also required under the WLQCL framework and are therefore beneficial for maintaining WHO-listed status.

In addition, for official (governmental) quality control laboratories, WLQCL listing requires that the regulatory system to which the laboratory belongs has achieved maturity level 3 for the laboratory function, as assessed through the WHO Global Benchmarking Tool (5). More broadly, the regulatory authority, as the leading institution within the national regulatory system of medical products, is encouraged to strive towards maturity level 3 across all regulatory functions.

The assessment procedure established by WHO is based on the following principles:

- evidence that the laboratory provides or is committed to offering quality control services for pharmaceutical products to United Nations agencies and their partners, national authorities of WHO Member States, and procurement agencies serving national authorities and United Nations agencies;
- a comprehensive understanding of the laboratory's documented quality management system and testing activities, including the use of recognized pharmacopoeias (for example, *The International Pharmacopoeia*);
- evaluation of submitted documentation;
- compliance with WHO-recommended quality standards for QCLs, that is, good practices for pharmaceutical quality control laboratories (1), including, where applicable, good practices for pharmaceutical microbiology laboratories (2), as well as the relevant parts of good manufacturing practices (3) and of the *WHO guideline on the implementation of quality management systems for national regulatory authorities* (4);
- monitoring of performance of WLQCLs and their compliance with WHO requirements.

WHO invites the national regulatory authority having regulatory oversight over a laboratory participating in the WLQCL procedure to join as an

observer in the audit of the laboratory's compliance with WHO-recommended standards for QCLs. WHO recommends that laboratories expressing an interest in participating in the WLQCL procedure inform the regulatory authority of the country in which they are established as well as relevant networks (for example, the European Network of Official Medicines Control Laboratories) of their submission for designation as a WLQCL.

2. Steps of the procedure

WHO requires information related to the activities of, and quality control of pharmaceutical products in, laboratories interested in being assessed under this procedure. Interested QCLs should submit the information about their activities as requested by WHO (see point 2.2 below). In addition to the evaluation of the information submitted, a site audit (or audits) may be performed.

If, due to insufficient resources and time constraints, WHO has to set priorities in the assessment of interested laboratories, then priority will be given to QCLs in areas where United Nations agencies and their partners, national authorities of WHO Member States, and procurement agencies serving national authorities and United Nations agencies identify the need for testing of the quality of pharmaceutical products.

Applications from laboratories that belong to or are affiliated with a manufacturer of pharmaceutical products, particularly those that have an interest in having one or more of their products prequalified by WHO or whose products are already prequalified by WHO, may be given lower priority or may not be evaluated at all.

WHO reserves the right to terminate the quality assessment of a laboratory when (a) the laboratory is not able to provide, or fails to provide, the required information; (b) when the information supplied is inadequate to complete the quality assessment effectively; (c) when the laboratory fails or refuses to collaborate during WHO audits; or (d) when the laboratory is unable to implement corrective actions that WHO may require within a specified time period.

2.1 Publication of invitation for expressions of interest

WHO will publish an invitation to QCLs to submit an expression of interest to participate in the WLQCL procedure. Such an invitation will specify the scope of quality control testing that is subject to WLQCL listing and will be published widely, that is, on the WHO website and possibly also through other media, such as the international press. The invitation will be open and transparent, inviting all interested QCLs to submit an expression of interest for WLQCL listing.

2.2 Submission of expressions of interest and laboratory information

Each interested laboratory should provide the WHO focal point indicated in the invitation for expressions of interest with:

- a cover letter expressing interest in participating in the WLQCL procedure;
- evidence that the laboratory provides, or is committed to offering, quality control services for pharmaceutical products to United Nations agencies and their partners, national authorities of WHO Member States, and procurement agencies serving national authorities and United Nations agencies;
- the relevant laboratory information.

WHO will record the receipt of the expression of interest from each laboratory in a register. If the laboratory has documented its quality system as a quality manual, this can be submitted, provided that it is supplemented with the information required for the laboratory information file (see below) that is not provided in the quality manual.

The information should be submitted as described in the *WHO Guidelines for preparing a laboratory information file* (6) and should cover the areas listed below, which are based on and complement the content of the guidelines:

- general information on the laboratory, including the activities proposed for WLQCL listing and, where applicable, activities related to the testing of vaccines, biologicals or advanced therapy medicinal products;
- accreditation or certification status, if applicable;
- quality management system implemented, and inspections and external audits performed in the laboratory;
- risk management;
- performance management;
- crisis management;
- communication management;
- participation in proficiency testing schemes or collaborative trials;
- internal audits;
- control of data, documentation and records;
- personnel;
- premises;

- equipment with qualification procedures;
- reagents, reference substances and reference materials;
- subcontracting of testing (where applicable);
- handling of samples;
- validation and verification of analytical procedures;
- evaluation and reporting of results;
- investigation of atypical results (out-of-specification, out-of-trend or out-of-expectation results);
- uncertainty of measurement;
- stability testing (where applicable);
- microbiological testing (where applicable).

Guidelines for the submission of expressions of interest and for the preparation and submission of the relevant information are available on the WHO website and will be sent to interested laboratories upon request.

2.3 Screening of submitted laboratory information

The *WHO Guidelines for preparing a laboratory information file* (6) will be used in an initial screening of the information supplied. The information will not be evaluated if it is not complete. In such cases the laboratory will be requested to provide additional information within a specified time. If the additional information is not received by the deadline, the application will be rejected.

2.4 Evaluation of the laboratory information

Laboratory information that complies with the requirements set out in section 2.2 will be evaluated in accordance with a standard operating procedure established by WHO to ensure uniformity in evaluation of the information. The information will be evaluated against the WHO-recommended quality standards for QCLs, namely good practices for pharmaceutical quality control laboratories (1), including, where applicable, good practices for pharmaceutical microbiology laboratories (2), as well as the relevant parts of good manufacturing practices (3) and of the *WHO guideline on the implementation of quality management systems for national regulatory authorities* (4). The outcome of the evaluation shall determine whether the laboratory will be considered for a possible site audit.

A laboratory may submit the report of the inspection performed by a regulatory authority, or the report of an audit conducted by an accreditation body or another competent body, together with the laboratory's response to any observations. These inspections or audits should have been conducted in accordance with standards at least equivalent to those outlined above.

Based on a standard operating procedure established by WHO for review of external inspections and audits, if the laboratory is considered to be operating at an acceptable level of compliance with WHO-recommended standards, WHO may decide that it is not necessary to conduct a site audit.

2.5 Site audit

Depending on the outcome of the evaluation of the laboratory information, WHO may plan and coordinate audits of the laboratory to assess compliance with the applicable WHO-recommended quality standards described above.⁸ The audit will be performed by an auditor, or a team of auditors, having the relevant qualifications and experience in the field of quality control of pharmaceutical products.

External auditors will be appointed in accordance with WHO procedures and will act as temporary advisers to WHO. The external auditors must comply with the confidentiality and conflict of interest rules of WHO, as laid down in the relevant sections of this procedure. A WHO staff member will coordinate the team. The auditor or audit team will perform the audits and report on the findings in accordance with standard operating procedures established by WHO to ensure a standard harmonized approach.

A representative or representatives of the national regulatory authority having regulatory oversight over a laboratory participating in the WLQCL procedure will be invited to accompany the team as an observer.

With a view to coordinating audit activities, avoiding duplication and promoting information sharing without prejudice to the protection of any confidential and proprietary information of the laboratory in accordance with the terms of this procedure, WHO may disclose audit-related information to regulatory authorities of WHO Member States, United Nations agencies and the European Directorate for the Quality of Medicines and HealthCare.

2.6 Report and outcome of audit

The auditor or audit team will finalize a report describing the findings according to the established WHO standard operating procedure and format. The report will be communicated by WHO to the laboratory and a copy will be sent to the national regulatory authority having regulatory oversight over the laboratory.

If any additional information is required, or if corrective action has to be taken by the laboratory, WHO will postpone its decision on the acceptability of the laboratory concerned until the additional information has been evaluated, or the corrective action has been taken, and found satisfactory. If the decision cannot be made based on the information received, a follow-up audit will be performed.

⁸ Training modules can be found on the WHO prequalification website (<https://www.who.int/tools/health-products-regulation-and-prequalification-learning-catalogue>).

In the event of any disagreement between a laboratory and WHO, a standard operating procedure for the handling of such disagreements will be followed to discuss and resolve the issue.

As WHO is responsible for the quality assessment procedure, the ownership of the reports lies with WHO (without prejudice, however, to any confidential and proprietary information of the laboratory contained in this report). Thus, WHO shall be entitled to use and publish such reports subject always, however, to the protection of any confidential and proprietary information of the laboratory. “Confidential information” in this context means:

- confidential intellectual property, “know-how” and trade secrets (including, for example, programmes, processes or methods, unpublished aspects of trademarks and patents);
- commercial confidences (for example, structures and development plans).

Provisions of confidentiality will be contained in the letters exchanged between WHO and the laboratory, to be agreed upon before the evaluation of the information and site audit.

Notwithstanding the foregoing, WHO reserves the right to share the full reports with the relevant authorities of any interested Member State of the Organization and interested United Nations agencies. In such cases, the concerned QCL shall be informed.

2.7 Results of assessment

Once WHO requirements are met, and the laboratory is considered acceptable in principle for use by United Nations agencies and their partners, national authorities of WHO Member States, and procurement agencies serving national authorities and United Nations agencies (that is, it has been found to meet the WHO-recommended quality standards for QCLs), the laboratory at the specified site will be included in a list referred to as “List of WHO-listed quality control laboratories”.

Laboratories on the list will be considered to be able to test products in compliance with WHO-recommended quality standards for QCLs. Inclusion in the list does not, however, imply any certification by WHO of the laboratories (which is the sole prerogative of national authorities).

Before publication of its name on the list of WLQCLs, each laboratory will receive a letter from WHO informing it of the outcome of the quality assessment process for that particular laboratory.

A copy of this letter will be sent to the national regulatory authority of the country where the laboratory is located. The list of WLQCLs will be published on the WHO website and will specify the areas of expertise assessed and considered listed. The list will be updated whenever new relevant information is obtained.

In line with World Health Assembly resolution WHA57.14 of 22 May 2004, WHO will – subject to the protection of any confidential and proprietary information – publish open-access reports on the laboratories considered to meet WHO-recommended quality standards for QCLs. These reports will be published on the WHO website.

2.8 Monitoring of WLQCLs

Once the laboratory is included in the list of WLQCLs, it should inform WHO without delay about the implementation of any subsequent changes that may have an impact on the WLQCL listing of the laboratory (such as changes to facility, equipment or key personnel) and should submit an updated laboratory information file.

Each WLQCL will be re-evaluated at regular intervals (annually), or earlier if information indicating the necessity for re-evaluation is obtained by WHO.

To enable WHO to carry out re-evaluation, all WLQCLs are requested to submit a brief annual report on their activities. The report should cover all activities related to quality control of pharmaceutical products within the preceding year and should be submitted by the end of March of the subsequent year. The following items should be included in the report:

- a summary of services provided to United Nations agencies and their partners, national authorities of WHO Member States, and procurement agencies serving national authorities and United Nations agencies;
- the number of samples analysed, differentiating between compliant and non-compliant samples, including any out-of-specification or out-of-trend investigations;
- a list of analytical methods used, including testing parameters and reference documents;
- a summary of customer complaints concerning results of analyses performed by the laboratory;
- brief details of participation in proficiency testing schemes (organizing party, methods involved, outcomes and, if appropriate, corrective measures adopted);
- listing of inspections and audits performed by external parties, identifying the party and the scope of the inspection and audit;
- confirmation of the primary and secondary contact points;
- in the case that changes have been implemented that have an impact on the content of the laboratory information file, a summary of these changes should be included in the report and an updated laboratory information file should be attached.

WHO will conduct re-auditing of WLQCLs in accordance with standard operating procedures established by WHO. The frequency of such re-audits depends on WHO's assessment of the quality risk management factors described below. Normally, however, such re-audits will take place at least once every three years. The following factors will be taken into account when planning audits:

- major changes, for example, to premises, equipment or key personnel;
- the results of previous inspections or audits by WHO or another external party, and history of compliance of the laboratory with WHO-recommended quality standards;
- the outcomes of participation of the laboratory in proficiency testing schemes;
- number and significance of customer complaints;
- laboratory experience with testing of pharmaceutical products;
- WHO's experience with testing services provided by the laboratory.

WHO reserves the right to proceed with the re-auditing of a WLQCL at any time this is considered necessary based on information or complaints received by WHO. The national regulatory authority that has regulatory oversight over the laboratory will be invited to participate in the re-auditing as an observer.

WHO may suspend or withdraw a WLQCL from the list of WHO-listed quality control laboratories when there is evidence of non-compliance with the WHO-recommended quality standards for such laboratories or this procedure.

The re-evaluation of the WLQCL status of a QCL may not be prioritized if the laboratory has not, for a continuous period of more than three years, provided quality control services for pharmaceutical products to United Nations agencies and their partners, national authorities of WHO Member States, and procurement agencies serving national authorities and United Nations agencies. In such cases, WHO will request the QCL to provide evidence that such services had been offered, or commitments made to continue to offer such services, to the aforementioned entities. In addition, continued compliance with WHO requirements shall be verified before the laboratory can resume providing services.

2.9 Monitoring of complaints

Complaints concerning the results of analysis of pharmaceutical products performed by the WLQCL or concerning the service provided by the WLQCL that are communicated to WHO will be investigated in accordance with WHO's standard operating procedure. The national regulatory authority that has regulatory oversight over the laboratory will be invited to participate in the investigation of the complaint.

After conducting its investigation, WHO will provide a written report of the problem, which may, where appropriate, include recommendations for action to the laboratory under investigation and to the national regulatory authority having the regulatory oversight over the laboratory.

2.10 **Cost recovery**

WHO reserves the right to charge for the quality assessment procedure on a cost recovery basis.

2.11 **Confidentiality undertaking**

WHO will require any external auditors (acting as temporary advisers to WHO) to treat all information to which they gain access during the audits of the laboratory, or otherwise in connection with the discharge of their responsibilities in regard to the WLQCL procedure, as confidential and proprietary to WHO or parties collaborating with WHO in accordance with the terms set out below.

Such auditors will be required to take all reasonable measures to ensure that confidential information:

- is not used for any purpose other than the activities described in this document;
- is not disclosed or provided to any person who is not bound by similar obligations of confidentiality and non-use as contained herein.

External auditors will not, however, be bound by any obligations of confidentiality and non-use to the extent they are clearly able to demonstrate that any part of the confidential information:

- was known to them prior to any disclosure by or on behalf of WHO (including by laboratories); or
- was in the public domain at the time of disclosure by or on behalf of WHO (including by laboratories); or
- has become part of the public domain through no fault of theirs; or
- has become available to them from a third party not in breach of any legal obligations of confidentiality.

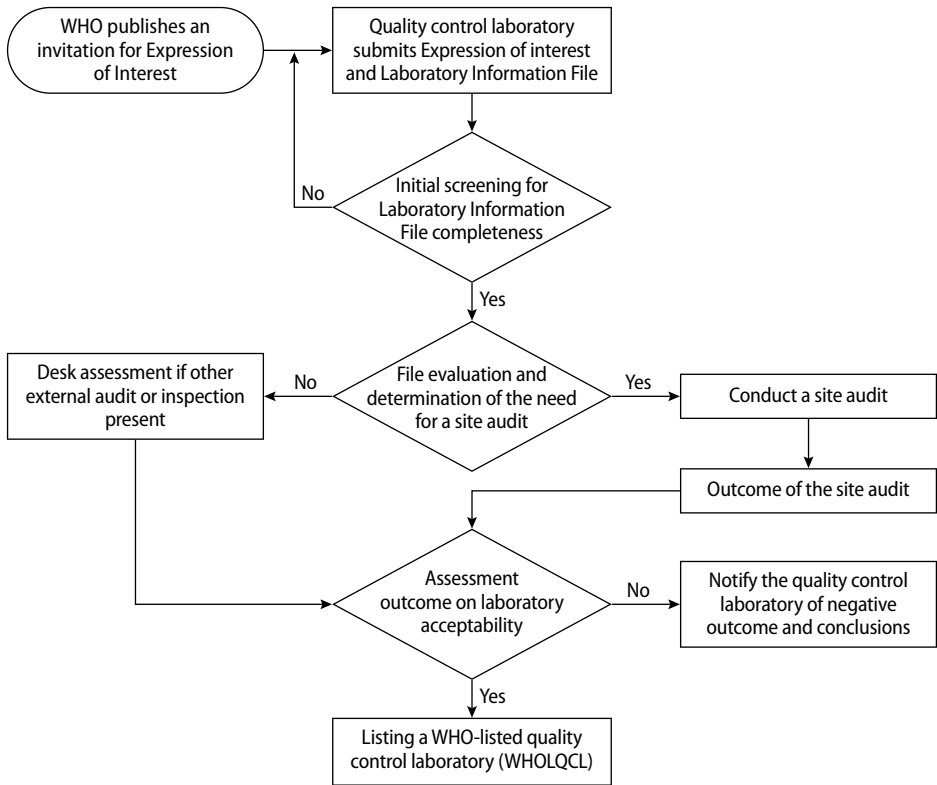
2.12 **Conflict of interest**

Before undertaking the work, each external auditor will also (in addition to the above-mentioned confidentiality undertaking) be required to sign a declaration of interest. If, based on this declaration of interest, it is felt that there is no risk of a real or perceived conflict of interest (or it is felt that there is only an insignificant or irrelevant conflict of interest), and it is thus deemed appropriate for the auditor in

question to undertake this work, they will discharge their functions exclusively as an adviser to WHO. In this connection, each auditor is required to confirm that the information disclosed by them in the declaration of interest is correct and complete, and that they will immediately notify WHO of any change in this information.

All external auditors furthermore agree that, at the laboratory's request, WHO will advise the laboratory in advance of the identity of each auditor and the composition of the team performing the site audit and provide curricula vitae of the external auditors. The laboratory then has the opportunity to express possible concerns regarding any of the external auditors to WHO prior to the visit. If such concerns cannot be resolved in consultation with WHO, the laboratory may object to an external auditor's participation in the site visit. Such an objection must be made known to WHO by the laboratory within 10 days of being notified of the proposed team composition. In the event of such an objection, WHO reserves the right to cancel its agreement with the auditor in question and the activities to be undertaken by that auditor, in whole or in part.

Fig. A5.1
Flowchart of WHO listing of quality control laboratories



References

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